

Ayush Chachan

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EDUCATION

Indian Institute of Technology, Delhi

July 2016 – June 2022

Dual Degree (B. Tech and M. Tech) in Biochemical Engineering and Biotechnology

CGPA - 8.905/10.000

EXPERIENCE

UPSC Civil Services Preparation |

Oct 2023 – Oct 2025

- Prepared for UPSC CSE with **Mathematics Optional**, involving sustained self-study and structured learning.

Helloverify India Private Limited | *Software Engineer*

July 2022 – Mar 2023

- Built scalable REST APIs using **Django REST Framework**, improving client onboarding speed and efficiency.
- Engineered **Selenium**-based web scrapers to automate data extraction, reducing the time by **82% per request**.
- Implemented **Celery**-based **schedulers** for asynchronous email and phone alerts, ensuring timely client updates.

Indian Institute of Technology, Delhi | *Graduate Researcher (M.Tech Thesis)*

May 2021 – June 2022

- Designed **statistical algorithms** for bioreactor data validation using mass balances and regression models.
- Applied **error propagation analysis** and consistency testing to identify anomalous and unreliable process data.
- Developed Python routines for **least squares estimation** and **chi-square testing** in over-determined systems.

SCHOLASTIC ACHIEVEMENTS

IIT Delhi BOSS Award (2022) - Conferred for best experimental and design-based research in the **M.Tech project**.

APPLICATION & INDUSTRY-SCALE SYSTEMS

Large-Scale Funnel Analytics Optimization | *Java, Binary Protocol*

- Re-architected a self-join-intensive analytics query into a **user-level aggregation model**, reducing complexity from **$O(n^2)$ to $O(n)$** and lowering BigQuery slot consumption by **80%**.
- Developed scalable **BigQuery JavaScript UDFs** operating on aggregated user event arrays using **binary-search-based** sequential evaluation.
- Built a fixed-schema ingestion pipeline using **Protobuf**, reducing network payload size by **40%** and improving serialization throughput by **2×**.
- Containerized services with **Docker** and automated deployments using **Kubernetes** and **GitLab CI/CD**.

Kafka-Like Log-Structured Messaging System | *Java, Distributed Systems*

- Built from scratch a **high-throughput append-only log** inspired by Kafka for large-scale event ingestion.
- Implemented **log segmentation**, **offset-based indexing**, and **sequential disk I/O** to optimize producer appends and consumer reads.
- Designed custom **Producer and Consumer abstractions** with independent offset tracking, deterministic ordering, and replayability.
- Modeled core messaging primitives including **topics**, **partitions**, **offsets**, and **immutable log segments** without using Kafka libraries.

Mini Spring MVC Framework with Custom HTTP Server | *Java, Networking, Concurrency*

- Implemented a **custom HTTP server from scratch**, parsing raw TCP byte streams and handling core **HTTP/1.1 request semantics**.
- Designed a **thread-pooled execution model** using **ExecutorService** to enforce thread limiting and backpressure.
- Built a lightweight **IoC container** with custom annotations for component scanning, bean registration, and dependency injection.
- Implemented a **Dispatcher and Controller Method Registry** enabling a complete Spring-like MVC request pipeline.

Unix-like Shell Implementation | *C, UNIX System Calls*

- Developed a Unix-style shell using **system calls (fork, exec, wait)** with **zombie process reaping** and robust error handling.
- Implemented **background (&), serial (&&), and parallel (&&&)** execution with custom **signal handlers**.

Monte Carlo Simulation | *Disjoint Sets, Union Find, Java*

- Modeled 2-D lattices using **graphs** and estimated percolation thresholds via **Monte Carlo simulation**.
- Optimized runtime using the **Disjoint Set (Union-Find)** data structure with path compression.

K-Dimensional Tree | *Java, BST, K-Nearest Neighbour Search*

- Implemented a **2D KD-Tree** for indexing points in a **K-dimensional space**.
- Supported Insert, Search, **Range Search**, and **Nearest Neighbour** queries in $O(\log^2 n)$ time.

Integer Arithmetic and Irrationals | *Karatsuba Multiplication, Newton's Method*

- Implemented **high-precision integer multiplication and division** using **Karatsuba's algorithm**.
- Computed irrational numbers up to **one million digits** using **Newton's Method** in $O(\log^2 d)$ time.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, Java Script, C/C++, Bash (Shell Scripting).

Frameworks & Libraries: Spring Boot, Hibernate/JPA, Django REST Framework, Celery

Databases & Storage: MySQL, PostgreSQL, Redis.

Web & API Development: REST APIs, Microservices, JSON, HTML5, CSS3.

Developer Tools: Git, Maven, Amazon Q, Bash, .

Data Analysis: Jupyter, NumPy, Pandas, Excel (Data Visualization).